

London Underground “What-If” Simulation

Requirements Document

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the London Underground Simulation Project. The system will simulate the London Underground network using real Transport for London (TfL) data to allow users to visualise routes, analyse disruptions and test network resilience through interactive scenario planning.

1.2 Scope

The system will provide an interactive, web-based simulation of the London Underground network. It will integrate real-time service data, support shortest-path routing, visualising network activity and allowing manual disruption testing.

The system does not handle fare calculation, ticketing, passenger accounts or real TfL operational control. It is intended for visualisation, routing and disruption analysis only.

2. System Overview

The system will consume TfL API data to construct a graph representation of the Underground network. Stations are represented as graph nodes, while connections between stations are represented as edges. A pathfinding algorithm will calculate optimal routes between selected stations, and the frontend displays the network on an interactive map.

Users can either observe live network conditions or enable What-If mode to manually close stations and lines. These simulated disruptions are applied locally and do not alter live TfL data.

3. User Classes

User Class	Description
Commuter	A casual user exploring journeys and disruption effects.
Transport Analyst	A user analysing network behaviour under disruption.
Plannner	A user testing hypothetical closure scenarios.
Academic Assessor	A user reviewing system correctness and implementation quality.

4. Operating Environment

- Client – Modern web browsers such as Chrome, Edge, Firefox and mobile browsers.
- Backend Runtime – Next.js.
- Database – Neo4j graph database.

- External Data Source – TfL Unified API.
- Network – Internet connectivity is required for live TfL data.

5. Functional Requirements

FR-01 Network Data Integration

The system shall retrieve station and stop hierarchy data from the TfL API.

FR-02 Station Hierarchy Mapping

The system shall represent stations as parent hubs with child nodes for platforms and entrances.

FR-03 Graph Construction

The system shall construct a connected graph representing all stations and lines.

FR-04 Route Calculation

The system shall calculate the shortest route between two selected stations.

FR-05 Travel Time Display

The system shall display estimated travel times for calculated routes.

FR-06 Real-Time Status Updates

The system shall retrieve live line status information and update the network accordingly.

FR-07 Dynamically Re-Routing

The system shall automatically recalculate routes when disruptions occur.

FR-08 Interactive Map

The system shall display the Underground network on an interactive map.

FR-09 Agent Simulation

The system shall simulate trains or passenger flows moving along network routes.

FR-010 Scenario Planning

The system shall allow users to manually close or reopen stations and lines.

FR-011 Visual Disruption Indicators

The system shall visually highlight closed or disrupted stations and lines.

FR-012 Crowd Simulation

The system may simulate overcrowding at stations based on capacity data.

6. Non-Functional Requirements

6.1 Performance

- System shall calculate routes within 2 seconds.
- Map interactions shall remain responsive under normal usage.

6.2 Reliability

- The system shall continue operating if the TfL API is temporarily unavailable using cached data.

6.3 Usability

- Interface shall be intuitive for first-time users.
- All controls shall be clearly labelled.

6.4 Scalability

- System shall support additional transport lines in the future.

6.5 Maintainability

- Codebase shall follow modular architecture principles.

6.6 Portability

- The system shall run on modern web browsers and phone browsers intuitively.

7. Assumptions and Dependencies

- TfL Unified API remains accessible and stable.
- TfL data accuracy is dependent on the external API provider.
- The London Underground topology does not change frequently.
- Users have internet access for live data features.
- The Neo4j database is available when the application requires graph data.